THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A device that provides the electrical energy to drive a current in a circuit is a: 1

(a) wire **(b)** bulb **(c)** switch **(d)** cell or battery

2. The heating element of an electric heater is made of: 1

(a) nichrome **(b)** plastic **(c)** iron **(d)** copper

3. A zigzag line in a circuit diagram represents: 1

(a) a resistor **(b)** a cell **(c)** a switch **(d)** a bulb

4. The symbol of a switch in the open (OFF) position shows: 1

(a) a cell **(b)** a closed loop

(c) a bulb **(d)** a gap in the circuit

5. A fuse wire has: 1

(a) no resistance **(b)** magnetic property

(c) high melting point **(d)** low melting point

6. If a bulb does not glow in a circuit, a possible reason is: 1

(a) a broken connection or a fused filament **(b)** the wires are copper

(c) too much light in the room **(d)** the cell is too big

7. In a closed circuit the current flows: 1

(a) only inside the cell

(b) in no direction

(c) from negative to positive only outside

(d) from the positive terminal through the circuit to the negative terminal of the cell

8. The filament of an electric bulb is made of: 1

(a) aluminium **(b)** lead **(c)** tungsten **(d)** copper

9. Which wire will heat up more for the same current? 1

(a) a thin nichrome wire **(b)** a thick copper wire

(c) a thick silver wire **(d)** a thick aluminium wire

10. Which of these is a good conductor of electricity? 1

(a) rubber **(b)** plastic **(c)** dry wood **(d)** copper

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is an electric circuit? 2

12. Why is the filament of a bulb made of tungsten? 2

13. Why do we use nichrome for heating elements? 2

14. Differentiate between a conductor and an insulator with one example each. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Draw in words a circuit with a cell, bulb and switch and explain how the bulb glows. 3

16. Explain how a fuse protects a circuit. 3

17. Write three uses of the heating effect of current. 3

OR

Explain why a bulb does not glow when the switch is open. 3

18. State three safety precautions while using electricity. 3

OR

Why do we need to use switches in a circuit? Give two reasons. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe Oersted's experiment with a labelled set-up and give its conclusion. State the effect of reversing the current. 5

OR

Explain the use of electromagnets in cranes and for separating iron. Why is soft iron used? 5

20. Describe how to make an electromagnet and test its strength. Mention factors that affect it. 5

OR

Explain how the thickness and the material of a wire affect heating, giving comparisons. 5

END OF QUESTION PAPER — Total: 40 Marks

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